

Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Vopak's 2026 Liquid Fuels Outlook shows sustained import volumes, with growth from accessible customers (cost × service)

Agenda	SCR	What the evidence shows	So what / story verdict	Deep dives
Market baseline	S1	Diesel drives imports; sales reports conflict 2024 net petrol/diesel imports: 13.078 bn L. Production, stocks, sales coverage and port allocation remain open.	Supported: imports are material. Sustained future volumes remain a forecast case; the matched balance is open.	3 / 6
	S2	Demand sits around Vopak's footprint GP, KZN and WC account for 68.6% of 2022 sales. Historical volumes were broadly steady; later provinces are estimates.	Supported: concentrated demand. Focus on Gauteng/KZN; sales totals do not establish accessible customers.	4 / 5
Market changes	S/C3	Demand drivers differ; L/M/H cases cover all key levers Power, freight, fleet, prices, supply and customer access need explicit assumptions; fuel effects remain uncalibrated.	Partly supported: diesel generation has fallen. Baseline, ample and tight worlds still need sourced assumptions and tested fuel/customer-volume effects.	7 / 9
	C4	Supply change makes import needs conditional Capacity has declined; redevelopment is conditional. Secunda/Natref product output and future cases remain open.	Supported: refining capacity fell. Plant futures are conditional; capacity alone does not determine imports.	8 / 9
	C5	Alternative gateways challenge the same inland market Matola and Walvis Bay have storage; their competitiveness depends on full costs, route capacity and access.	Open: gateway competition. No route winner without comparable cost, service and customer access.	10 / 11
Vopak outlook	R6	Market share of Vopak in Dbn and Lesedi Vopak has an established storage footprint; its share of customer volumes remains to be quantified.	Footprint established; volume share unquantified. Unique deliveries and customer access establish served volumes.	12 / 13 / 14
	R7	Potential scope for new investment? Prioritise Durban/Lesedi customer flows. Additional storage follows secured volumes and a usable-capacity gap.	Conditional: win customers first. Use existing assets; new storage needs secured flows and a usable-capacity gap.	15 / 16 / 17

Note: S = Situation C = Complication R = Resolution | Every Henry storyboard prompt has a response on pages 20–22; open items remain open.

Source: Partner SA storyboard; SARS / Stats SA / operator evidence, 6 Oct 2026. Remaining gaps labelled.

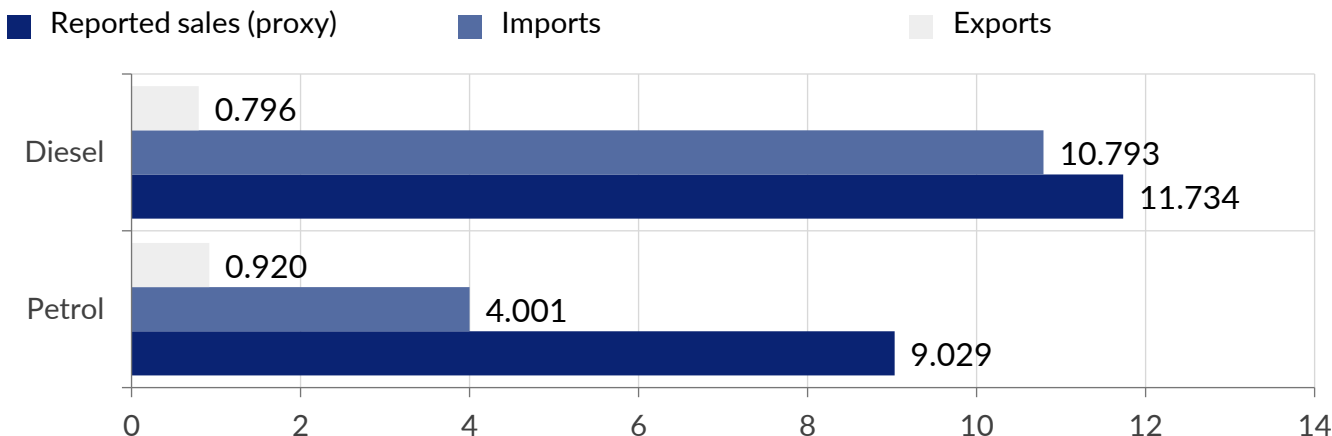
Strictly Confidential

S1. Diesel drives import volumes, while conflicting reports leave 2024 sales unresolved

National petrol/diesel | 2024, billion litres

Production + imports – exports – stock build = consumption

Reported sales are a consumption proxy; production and stock changes are missing.



Net imports: petrol 3.081, diesel 9.997; combined 13.078 bn litres.

Primary trade | SARS customs

Diesel imports 10.793 bn L.

Government rounds to 10.800; FIASA 14.793.

Use SARS; retain the source discrepancy.

2024 accounting remains open

Sales less net imports = 7.685 bn L.

Balancing requirement, not production.

Production, stocks and coverage unresolved.

What the evidence shows

01 | Customs supports the trade selection

SARS records diesel imports of 10.793 and exports of 0.796 bn litres in 2024. Net imports are 9.997; use customs as the primary trade source.

02 | Reported sales remain a proxy

FIASA reports petrol/diesel sales of 20.763 bn litres for 2024. Editions differ; this demand proxy remains unverified and is not imports plus exports.

03 | Production and stocks remain open

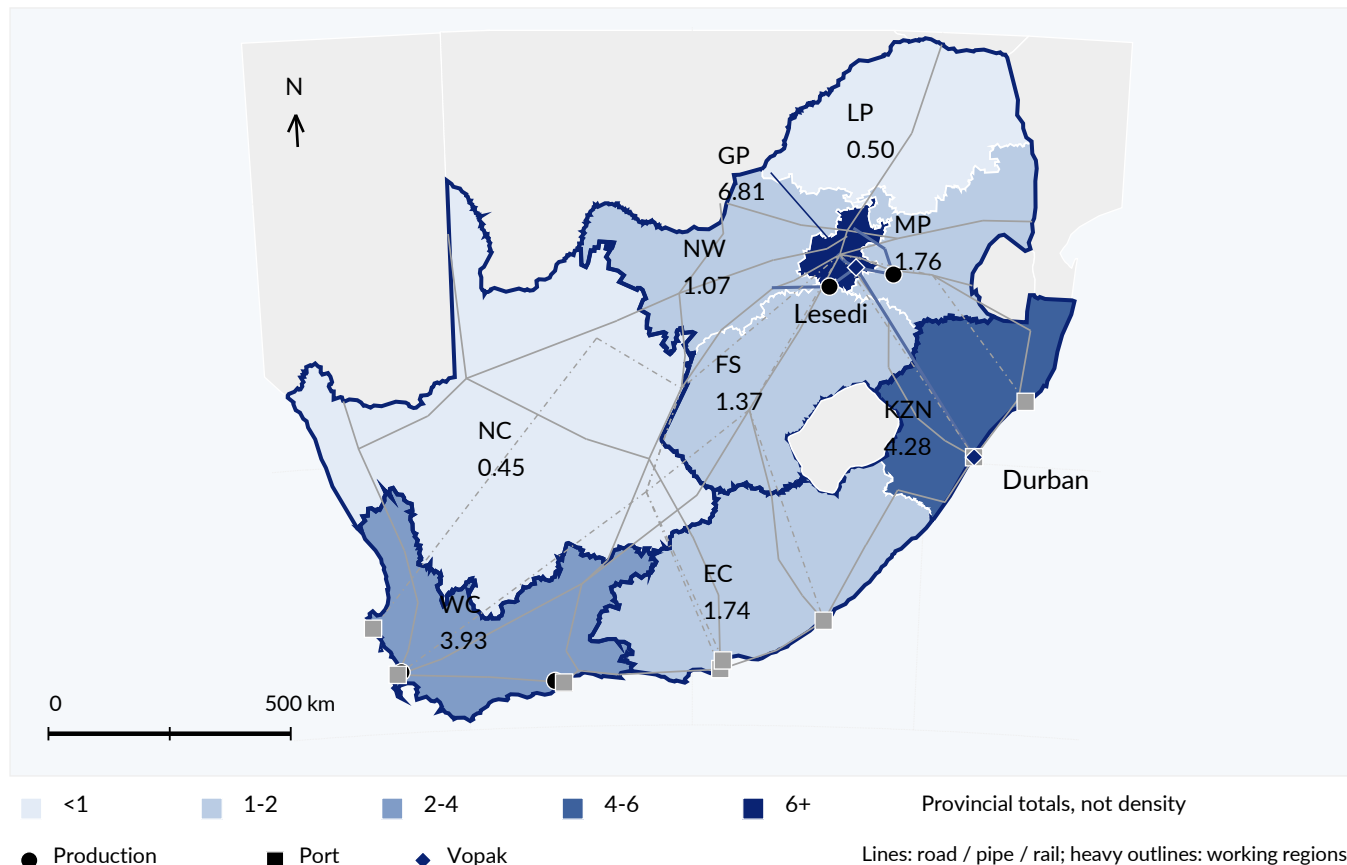
Production and stocks remain unresolved. The proposed 3-4 bn litre diesel gap is a hypothesis, not measured consumption.

To resolve next

Manish: rebuild the balance CSV from SARS; match product, year, actual output and stock changes. Nigel: agree the sales selection and obtain customer/port evidence.

S2. Three provinces account for 69% of reported fuel sales; the top two contain Vopak's storage facilities

Petrol + diesel sales by province | 2022, bn litres/year



What the evidence shows

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Four reporting regions

Eastern coast: EC/KZN. Inland: GP/FS/LP/MP/NW. Western coast: WC. Other: NC. Boundaries are reporting groups.

03 | Sales locate the historical market

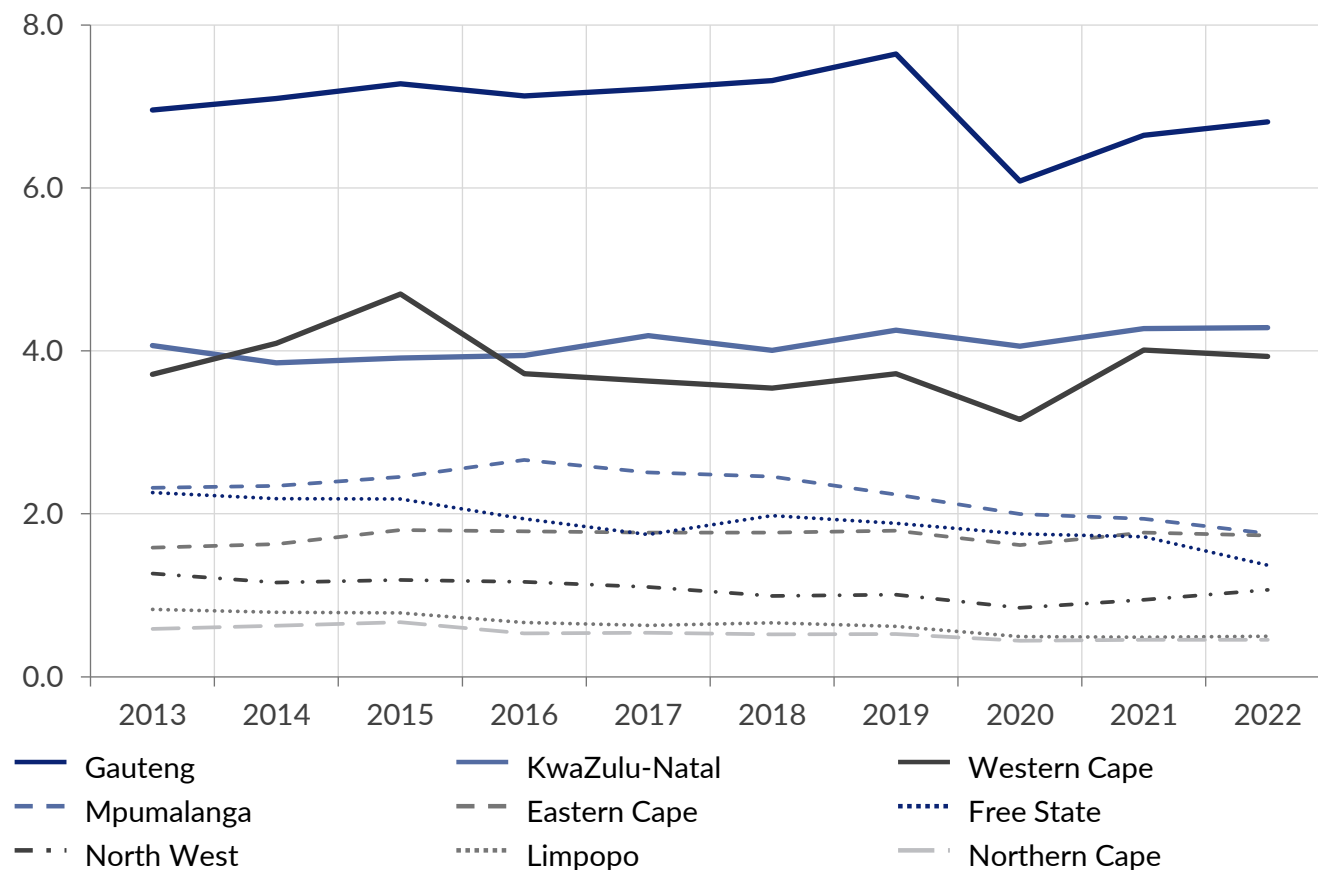
Colours show provincial sales totals, not density or customer catchments. Only Durban and Lesedi are labelled; competitor detail is on pages 13 and 15.

To resolve next

Nigel: confirm customer markets for Durban and Lesedi. Manish: retain the observed 2022 map; present later estimates separately with national-source flags.

S2. Reported fuel sales were broadly steady through 2022, despite the pandemic dip

Annual provincial sales | 2013–2022, bn litres/year



Source: DMPR provincial sales 2013–2022; Manish held-share 2024 petrol estimates (FIASA total unverified).

What the evidence shows

01 | Gauteng remains the largest market

The reported series places Gauteng above the other provinces throughout 2013–2022. Volumes differ across the smaller markets.

02 | Later provinces are estimates

2024 petrol estimates: Gauteng 3.47, KZN 1.47, Western Cape 1.41 bn L. These hold 2022 shares; no later observed provincial sales were found.

03 | Historical differences are traced

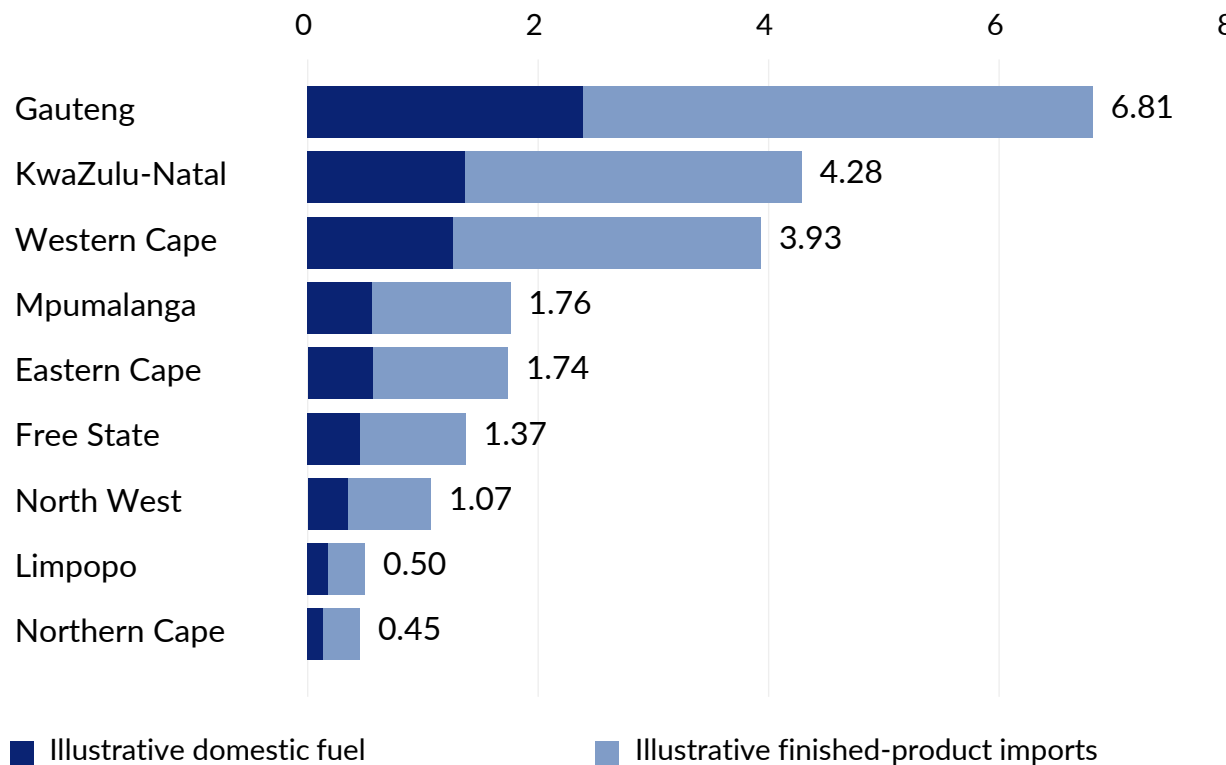
The analyst branch corrects the 2013 parser. Sources differ in 2014/2015; 2018 district coverage is incomplete. Retain these flags.

To resolve next

Manish: reconcile source selections and retain estimate labels. Nigel: decide whether estimated petrol volumes belong in the outlook; diesel booking shares are less stable.

S1. Provincial demand is known; its supply origin still needs evidence

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

Source: DMPR 2022 provincial sales totals; source mix from the existing illustrative national-balance CSV. Not observed origin.

Strictly Confidential

What the evidence shows

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

To resolve next

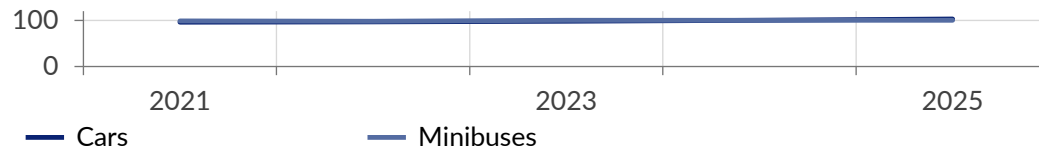
Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

S/C3. Activity trends show potential demand shifts; fuel used per unit of activity still needs validation

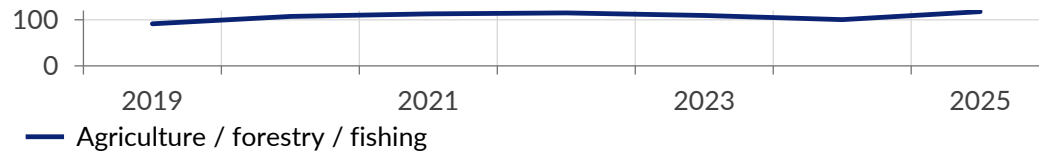
Potential demand levers | 2024 = 100; price panel uses nominal R/litre

Observed proxies, not fuel-volume effects. Activity axes 0–120; EV 0–400; prices 0–30 R/L.

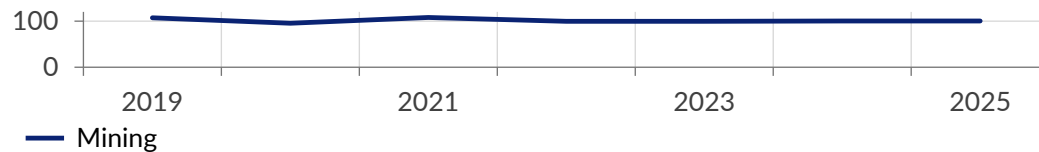
Passenger travel | all-fuel stock, Dec



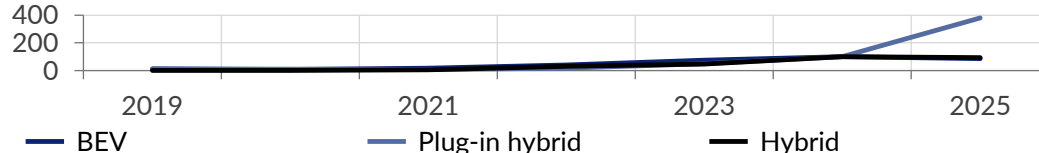
Agriculture | real value added, CY



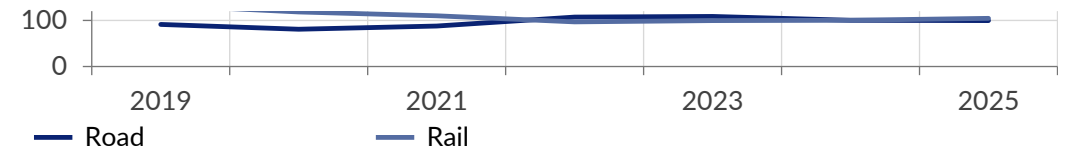
Mining | production volume, CY



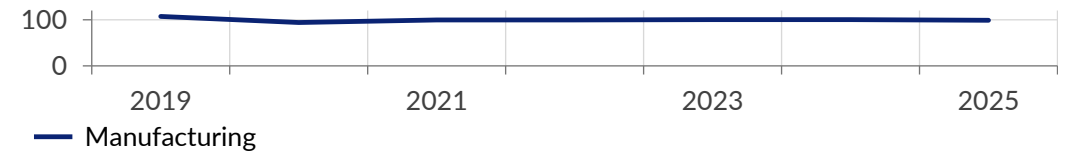
Electrification | annual new sales, CY



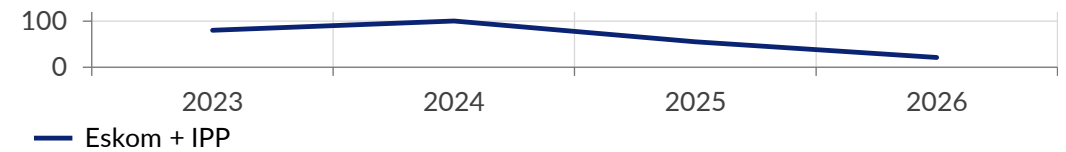
Freight demand | payload, CY



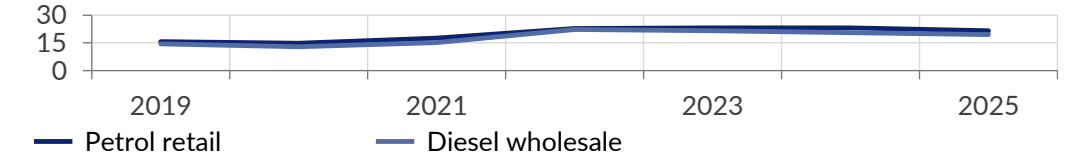
Manufacturing | production volume, CY



Diesel power | OCGT generation, FY



Price response | annual nominal R/litre



Note: Latest staged: activity Jul 2026; GDP Q2 2026; inland prices Oct 2026. Charts use complete years. Fuel-use rates need validation; FY ends March.

Source: NaTIS; Stats SA P0441 / P2041 / P3041.2 / P7162; Eskom; naamsa; DMPR / CEF. Snapshot manifest in notes.

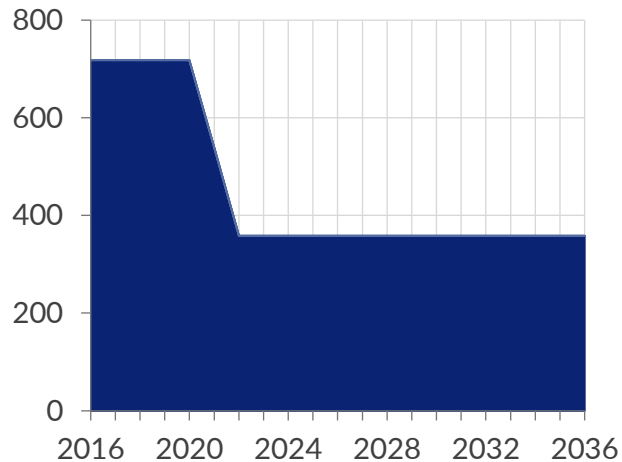
Strictly Confidential

C4. Refinery capacity has fallen; redevelopment remains conditional

Published footprint + illustrative scenarios | thousand bbl/day

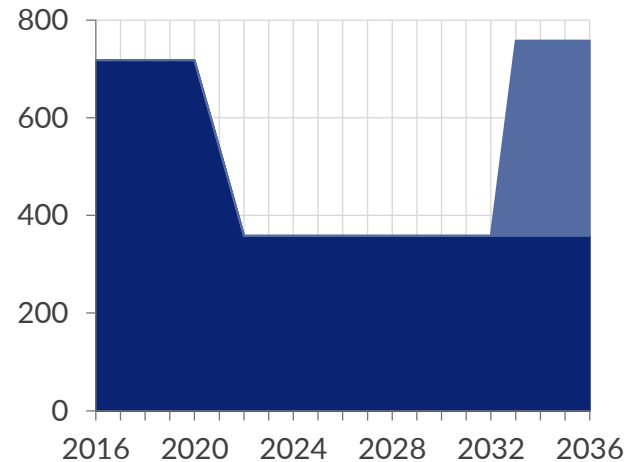
History 2016–2025 is sourced; 2026 onward is illustrative, not a forecast

A | No new refinery capacity



2036: 358 thousand bbl/day

B | Conditional CEF redevelopment



2036: 758 thousand bbl/day

Reported FY2024 output: Secunda 29.1; Natref Sasol share 17.8 million barrels.

What the evidence shows

01 | Capacity has fallen

Published capacity falls from 718 to 358 thousand bbl/day between 2020 and 2022. Redevelopment remains conditional.

02 | Reported output is different

FY2024 Secunda output is 29.1 million barrels; Natref is 17.8 for Sasol's share. Both cover all products and a year to June.

03 | A product bridge is needed

Match product yields, calendar years and downtime before calculating import effects. Capacity does not measure actual output.

To resolve next

Manish: match operator output to product/year and document Natref ownership basis. Henry: review technical mechanisms. Nigel: agree refinery scenario timing.

S/C3 / C4. L/M/H assumptions define coherent demand, supply and customer-access worlds

Scenario design | all key levers and assumptions

Input family	L/M/H settings to agree	Outputs and decision
Demand	Growth, activity, mileage, fleet efficiency/EVs, freight mode, power diesel use and price response	Petrol/diesel demand by year and region
Supply and trade	Plant output, yields, downtime, restart timing, import availability, exports and stocks	Domestic supply and required trade; residuals explicit
Customer access	Delivered price, service, feasible routes, rights and switching	Reachable and competitively accessible customer litres
Capacity and operations	Usable tanks, receipt/dispatch, inventory policy, peaks and spare space	Unique Vopak flows and any usable-capacity gap

Combine settings into coherent worlds

Baseline	Ample supply	Tight supply
Agreed reference assumptions	Supply and access comfortably meet demand	Demand tests available supply and delivery routes

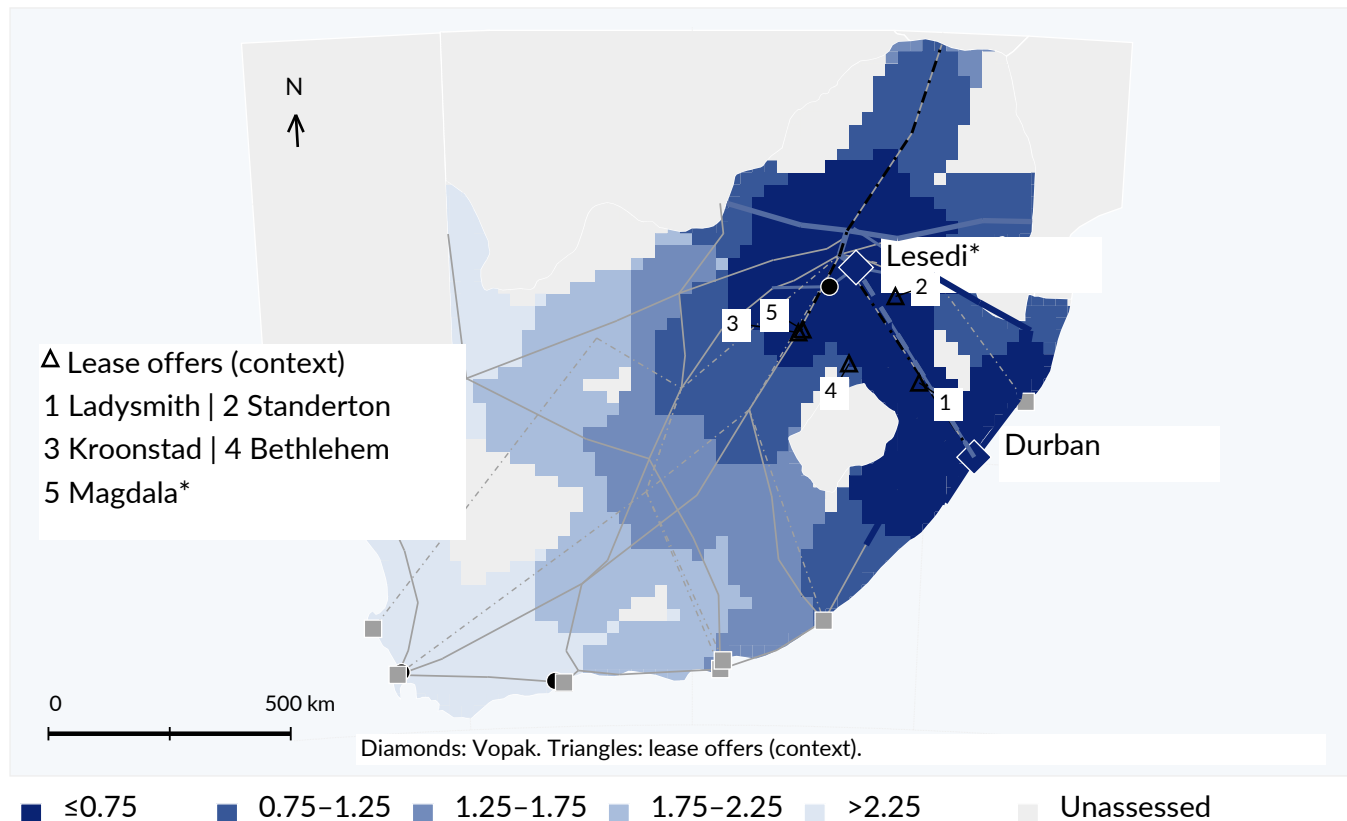
Note: History is observed; forecast cases are proposed. No new fuel forecast is shown until the assumptions, source selections and back-tests are reviewed.

Source: Authored forecast framework; observed series in reporting evidence snapshots. No new growth rates or fuel forecasts inferred.

Strictly Confidential

C5. Transport cost shapes the market Durban and Lesedi can reach

Illustrative road cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

What the evidence shows

01 | Darker means lower transport cost

Cells show example road cost from Durban or Lesedi. The rate is illustrative and excludes fuel price, storage and pipeline tariffs.

02 | Grey means unassessed

Grey areas have no assessed road connection under this illustration. They are not confirmed to be inaccessible.

03 | A cost surface is a first screen

Pipeline and rail lines are context. Route capacity, product compatibility and customer rights still determine serviceable demand.

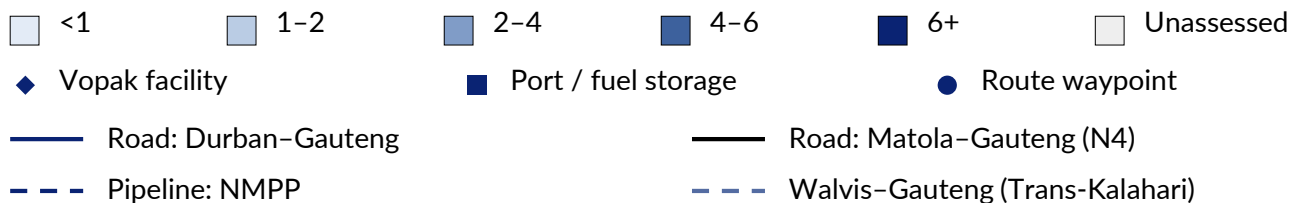
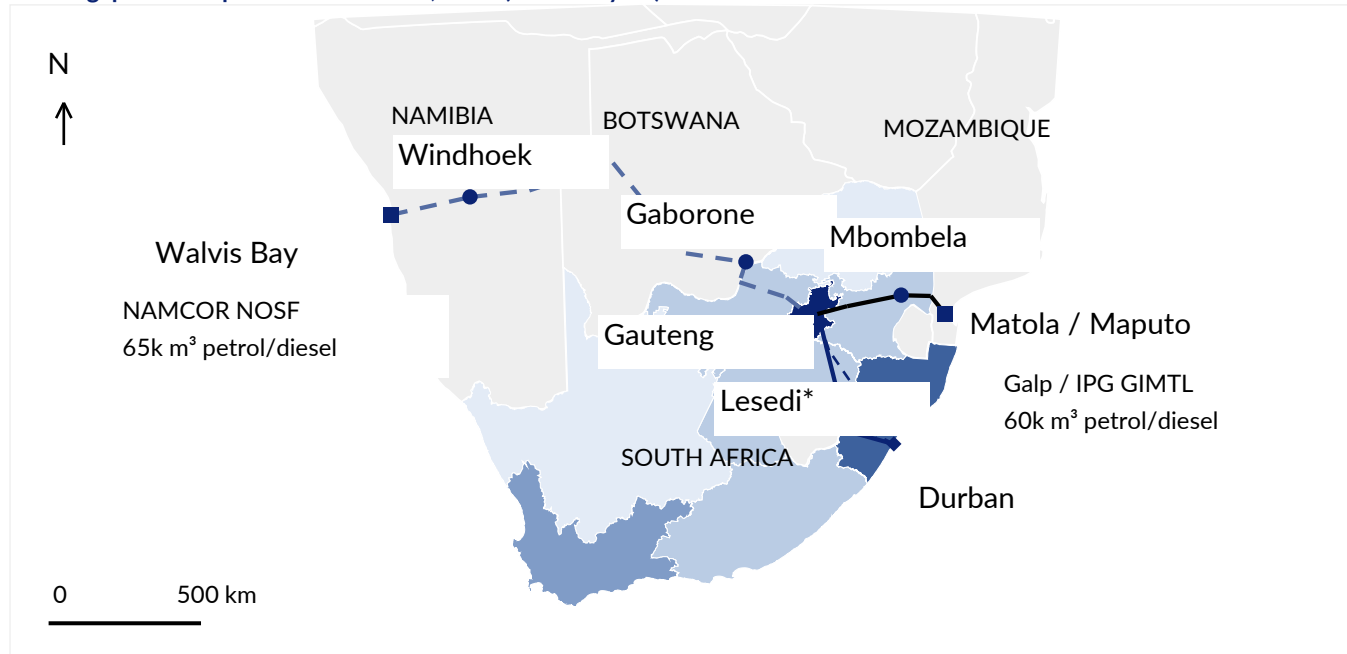
To resolve next

Nigel: select the first customer destination and cost threshold. Manish: replace example rates with dated route costs and access evidence.

C5. Compare Durban, Matola and Walvis Bay for the same inland customer

Durban, Matola and Walvis Bay | schematic corridor comparison

Shading: provincial petrol + diesel sales, 2022 (bn litres/year)



Note: Routes schematic; access unverified. Colour measures provincial totals, not density or Vopak share.
 Source: DMPR 2022 sales; NAMCOR NOSF; Galp GIMTL; WBCG / TRAC corridors; Natural Earth. Capacities and URLs in notes.
 Strictly Confidential

What the evidence shows

01 | Vopak has a South African footprint

Durban and Lesedi are highlighted against 2022 provincial sales. Colour shows sales totals; it does not establish Vopak market dominance.

02 | Matola has published fuel storage

Galp/IPG GIMTL lists 60,000 m³ of petrol/diesel tanks, plus 6,000 m³ LPG excluded here. N4 is the eastern inland comparison route.

03 | Walvis has storage and handling

NAMCOR NOSF lists 75,000 m³ in total, including 65,000 m³ petrol/diesel. Trans-Kalahari is the western inland comparator.

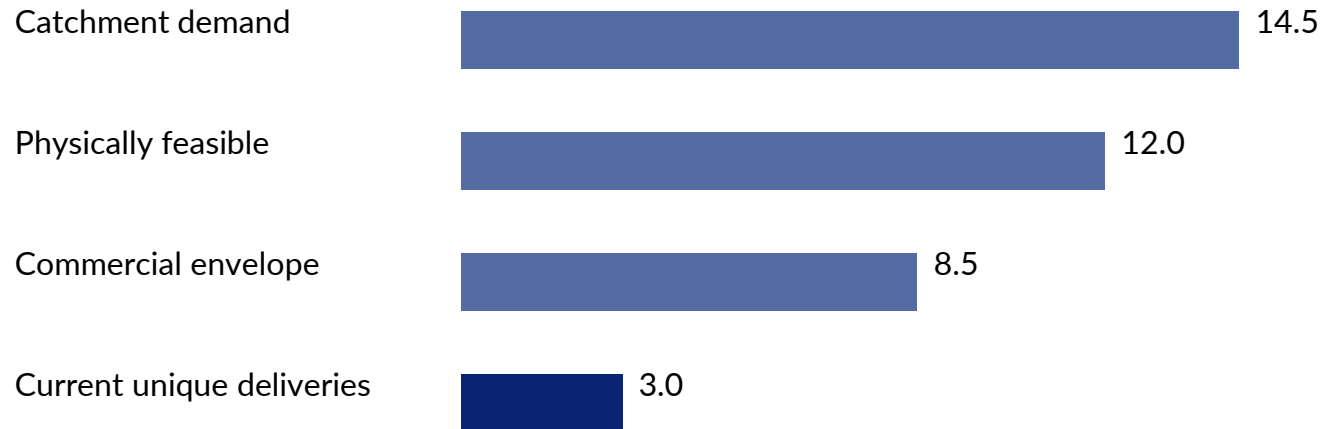
To resolve next

Manish: verify available tanks, customer access and full delivered costs. Source Namibia/Mozambique demand before extending the heatmap; grey means unassessed.

R6. Separate market demand, accessible volume and unique Vopak deliveries

Market screens | illustrative bn litres/year

Successive screens; these volumes are illustrative



Count Durban–Lesedi transfers once

Durban 2.8 + Lesedi 2.0 – shared transfer 1.8 = unique 3.0

Demand ≠ terminal receipts ≠ storage capacity. Incremental opportunity is on page 14.

What the evidence shows

01 | Count customer demand once

Example receipts of 2.8 at Durban and 2.0 at Lesedi include a shared 1.8 transfer. Unique demand served is therefore 3.0 bn litres.

02 | Accessibility precedes opportunity sizing

Example catchment demand 14.5 → physically feasible 12.0 → commercial envelope 8.5. Each screen requires evidence.

03 | Examples are separate from sales

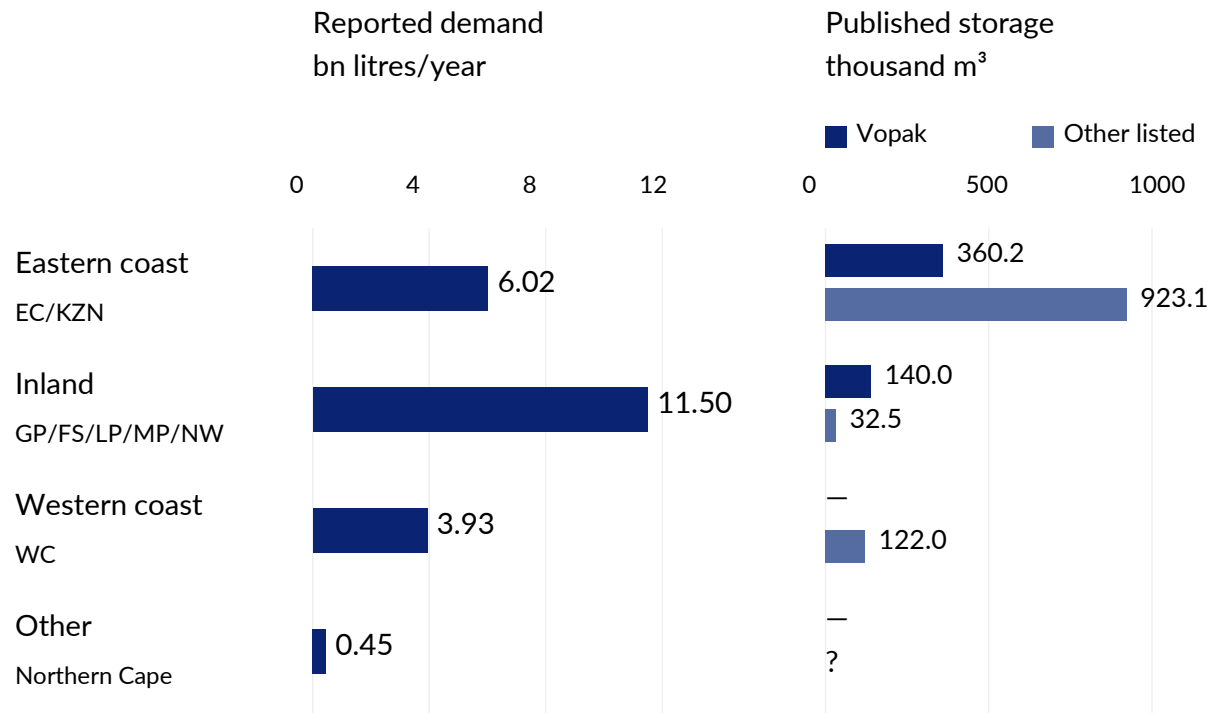
These market volumes illustrate the calculation. They are not the reported 2022 regional totals or actual Vopak deliveries.

To resolve next

Nigel: request customer deliveries, destinations and transfer records. Manish: calculate unique served demand before estimating additional access.

R6. Vopak's published tanks sit in the two largest regional demand markets

Historical demand and published tanks | separate scales



— no listed Vopak site ? capacity unknown

2022 total 21.90 bn L; petrol + diesel, jet excluded. Partial gross inventory; mixed products and missing Sasol/Transnet values. Tanks do not establish annual deliveries or market share.

[1] Vopak [2] Bidvest [3] Burgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

Strictly Confidential

Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.

Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].

Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Burgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

R6. Greater market penetration depends on reach, price, service and customer access

From total market to customer volumes | gates to test

- **Total market (TAM)**
Same product, year and customer geography
- **Infrastructure reach**
Feasible routes; compatible receipt and dispatch
- **Competitive price × service**
Full delivered cost and service versus alternatives
- **Commercial customer access**
Contracts, rights and ability to switch
- **Captured unique deliveries**
Additional customer litres; shared transfers removed

Today's served volumes + additional candidates that pass every gate

Current share and additional capture remain unquantified. Capacity, delivered cost and customer rights limit the addressable catchment.

Source: Existing illustrative catchment CSV; conditions authored for review. No actual Vopak share, cost threshold or forecast capture established.

Strictly Confidential

What the evidence shows

01 | Reach defines a candidate market

Routes locate potential customers. Receipt, dispatch and product constraints limit physical reach.

02 | Price and service determine competitiveness

Compare full delivered cost and service with alternatives for the same customer, product and period.

03 | Customer access determines capture

Test contracts and switching. Count unique final deliveries; keep unknown and failed gates explicit.

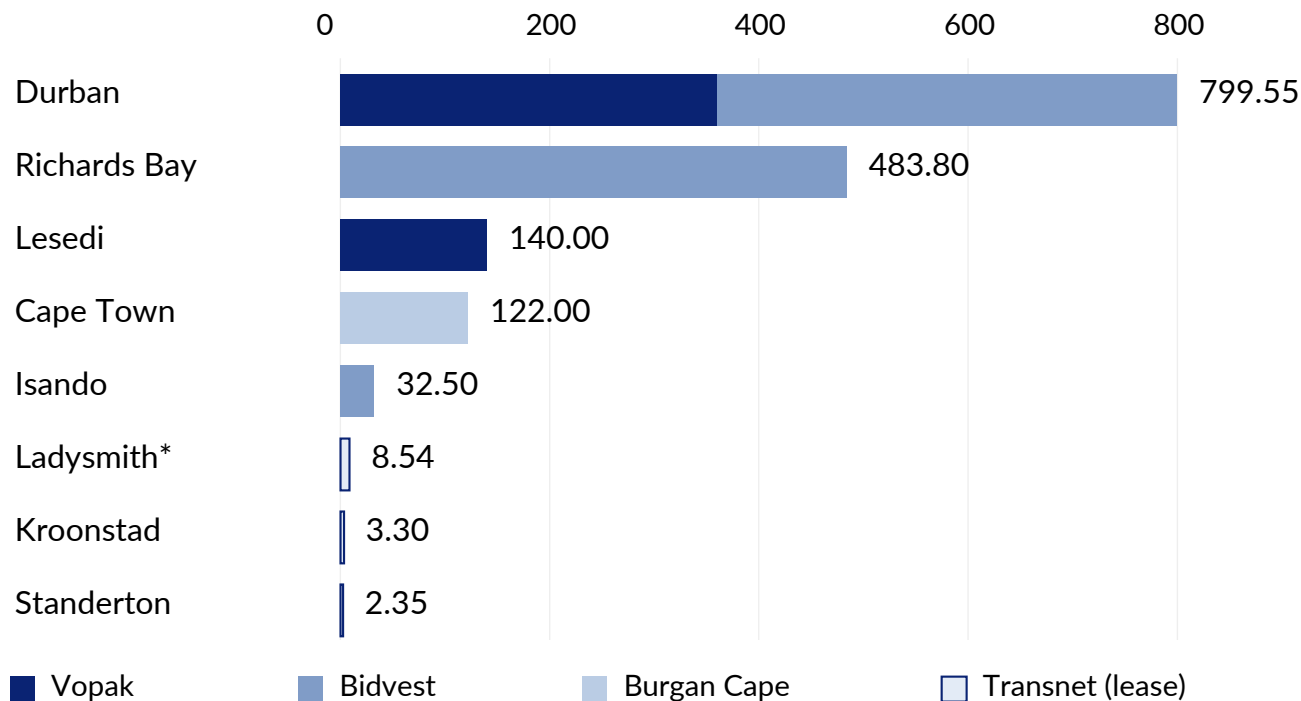
Next actions

- | | |
|---------------|--|
| Nigel | Agree destinations and delivered-cost thresholds. |
| Manish | Record passed, failed and unassessed customer-route gates. |

R7. Durban leads the published storage subset

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

What the evidence shows

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

To resolve next

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

R7. Prioritise customer markets, win on service and cost, then expand

Market outlook | priority hypotheses and conditional timing

			
Where to play	How to win	When to act	What must be true
Durban coastal customers First market to test	Competitive delivered cost; reliable receipt and dispatch; compatible tanks	Now: identify customers and test service using existing assets	Named customer, product and unique litres; cost threshold; usable tank and dispatch space
Lesedi inland customers First inland market to test	Reliable inland supply; competitive full-route cost versus other gateways	After route and customer validation: negotiate incremental flows	NMPP or alternative route access; receipt/dispatch capacity; customer rights and switching
Matola / Walvis Bay port-storage gateways Compare inland supply routes	Test storage, handling and transport to the same inland destination	During the inland comparison; pursue an opportunity only if access and economics are evidenced	Available compatible tanks; border/route feasibility; partner and customer access; full delivered cost

Investment gate

New storage follows secured incremental flows and a gap in usable capacity; the illustrative 5.5 bn litres does not establish a commitment.

Nigel: Choose the first customer/product and acceptable cost. Confirm the service proposition and commercial access.

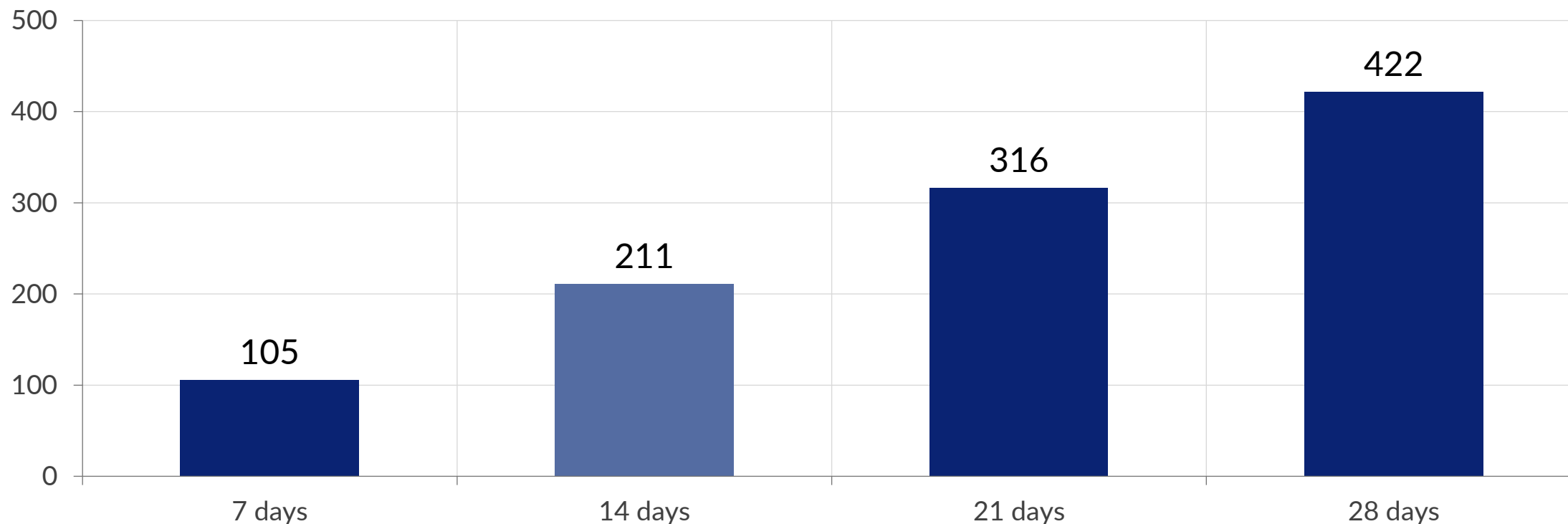
Manish: Build a customer-by-route evidence table: litres, cost, capacity, rights, timing and owner. Mark each gate passed, failed or unassessed.

Source: Authored market hypotheses; demand, gateway and storage evidence on pp.4, 10–15. Customer capture remains unverified.

R7. Inventory policy determines the working stock needed for additional flows

Working inventory, thousand m³

Illustrative: 5.5 bn litres/year; uniform daily flow



Inventory days

Working stock, not new capacity: allow for spare tanks, peaks, product segregation, tank heels and dispatch limits.

Source: Existing 5.5 bn L/year candidate example; authored inventory-day sensitivity, 6 Oct 2026. Not new-capacity sizing.

Appendix

The main story ends on page 17.

Document scope · Henry storyboard responses · supporting presentation palette

Use the supporting material to trace the evidence, assumptions and remaining work behind the outlook.

About this document

What it covers

South Africa petrol/diesel market baseline, historical demand and national supply/trade evidence.

Demand drivers, refinery alternatives and L/M/H scenario design for all key levers and assumptions.

Vopak customer opportunities through infrastructure reach, delivered price, service and commercial access.

Preliminary inventory sensitivities and a trace of the evidence needed to answer Henry's storyboard.

Week 1 working assessment, 6 October 2026. Observed data, estimates, conditional cases and illustrations are distinguished. Source gaps and proposed actions remain explicit; governance coverage is not data validation or business approval.

What it does not establish

A validated 2026 forecast or calibrated L/M/H scenario results; assumption ranges and back-tests remain work to do.

Measured Vopak market share, confirmed customer capture or verified operational catchments.

A closed national fuel balance, complete fuel-compatible storage census or confirmed spare tank capacity.

An investment recommendation, detailed engineering design or approved commercial business case.

S1 / S2 / S/C3 / C4. Henry storyboard answers | Market baseline

S | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Total SA demand development	Partial: reported 2024 sales are 20.763 bn L; coverage awaits reconciliation. Provincial history ends in 2022.	3 / 4 / 5
Eastern SA market within economic striking distance	Open: location and illustrative costs show reach, not accessible customer litres. Need destinations, tariffs and rights.	4 / 10 / 12 / 14
Greater diesel use in power generation	Partial: generation history is shown; recent OCGT use falls. Diesel-litres attribution still needs fuel-use evidence.	7 / 9
Rail to road	Partial: road/rail payload history is shown. Payload alone does not prove modal substitution or diesel effects.	7 / 9
Changes in fuel price	Partial: complete-year price history is staged. Travel response and real-price effects remain uncalibrated.	7 / 9
More fuel-efficient vehicles	Open: fleet stocks and new EV/hybrid sales do not establish fleet efficiency. Need fuel mix, mileage and intensity.	7 / 9
Slow GDP growth	Partial: GDP and sector activity evidence is available; growth-to-fuel effects require explicit assumptions.	7 / 9
Refinery closures	Partial: published capacity falls to 358 thousand bbl/day. Capacity is distinct from output and operating availability.	8
Challenges at Secunda	Partial: FY2024 output is 29.1 million barrels, all products. Availability and petrol/diesel yields remain open.	8 / 9
Closure of Moss gas	Open: plant-specific shutdown dates, operating status and lost product output need explicit source confirmation.	8 / 9
Growth in imports and where they enter	Partial: 2024 net imports are 13.078 bn L. Import-growth attribution, product-specific ports and origin remain open.	3 / 6

Source: Henry storyboard pp30–32; reporting evidence and explicit gaps, 6 Oct 2026. Responses do not establish closure.

S/C3 / C4 / C5. Henry storyboard answers | Market changes

C | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Road to rail; private-sector rail projection	Open: historical payload is available; private-sector service, timing and modal-shift assumptions need evidence.	7 / 9
End of loadshedding and spare power capacity	Partial: observed OCGT generation falls. Future dispatch, spare capacity and diesel demand remain conditional.	7 / 9
EV / HEV penetration projection	Open: sales uptake is shown, not fleet penetration. Separate BEV, PHEV and conventional hybrids; model turnover.	7 / 9
Effect of EV / HEV on litres per km	Open: require stock mix, mileage and fuel intensity. BEVs and hybrids have different fuel-use mechanisms.	7 / 9
Higher fuel prices and km travelled	Open: prices are sourced; mileage response and elasticity must be specified and tested before forecasting.	7 / 9
Mega refinery / SAPREF restart	Conditional: redevelopment capacity is a scenario, not sanctioned output. Confirm FID, ramp-up and product yields.	8 / 9
Secunda gas, coal and MRG diversion to 2032	Open: no liquid-fuel displacement is inferred. Need dated gas allocation, plant mechanism and product-output evidence.	8 / 9
Natref future?	Open: reported FY output is available; future ownership, availability and product yields require a dated case.	8 / 9
Durban storage / NMPP / Lesedi vs road and rail	Open: compare the same customer and product. Need full delivered costs, capacity, tariffs and service availability.	10 / 11 / 14
Transport from Matola and Walvis Bay	Partial: gateways, published storage and schematic routes are shown. Full costs, borders and customer access remain open.	11 / 14 / 16

Source: Henry storyboard pp30–32; reporting evidence and explicit gaps, 6 Oct 2026. Responses do not establish closure.

R6 / R7. Henry storyboard answers | Vopak outlook

R | Every prompt addressed; partial and open findings require the stated evidence

Henry storyboard prompt	Current answer / evidence gap	Deep dives
Market share of Vopak in Dbn and Lesedi	Open: published storage is an asset footprint, not served demand. Need unique deliveries, transfers and matched demand.	12 / 13 / 14
Potential scope for new investment?	Conditional: prioritise customer flows and service first. New storage requires secured incremental flows and a usable-capacity gap.	15 / 16 / 17

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com



Presentation palette | colours and HEX codes

Reference | colours used in this presentation

Core presentation colours

	Vopak blue #0A2373		Secondary blue #546CA2		Text / ink #000000		White #FFFFFF
	Unassessed / neutral #EEEEEE		Header divider #A0A0A0		Body divider #B8B8B8		Frame divider #B0B0B0

Sales-map colours | provincial totals, bn litres/year

	Below 1 #E3EBF6		1 to below 2 #BACCE4		2 to below 4 #809CC7		4 to below 6 #3D619D
	6 and above #0A2373		Map background #F5F8FB				

Other template theme colours

	Theme dark 2 #11151A		Theme light 2 #E7E6E6		Accent 3 #BDBEC1		Accent 4 #404040
	Accent 5 #767676		Accent 6 #D9D9D9				